MATH 206 - STAT 226 Probability and Statistics

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Stochastic Process

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Definition

A sequence of experiment is called "stochastic Process". That is let E_1, E_2, $\cdots$, E_n be n experiments followed by each other. Here, the problem is to determine the probabilities of a stochastic process. A tree diagram is used for this purpose.

Example

Given two boxes as follows:
Box I: 3 red and 2 white balls
Box II: 2 red and 3 white balls
First, a coin is tossed. If head occurs, select a ball from Box I, if tail occurs, select a ball from Box II.

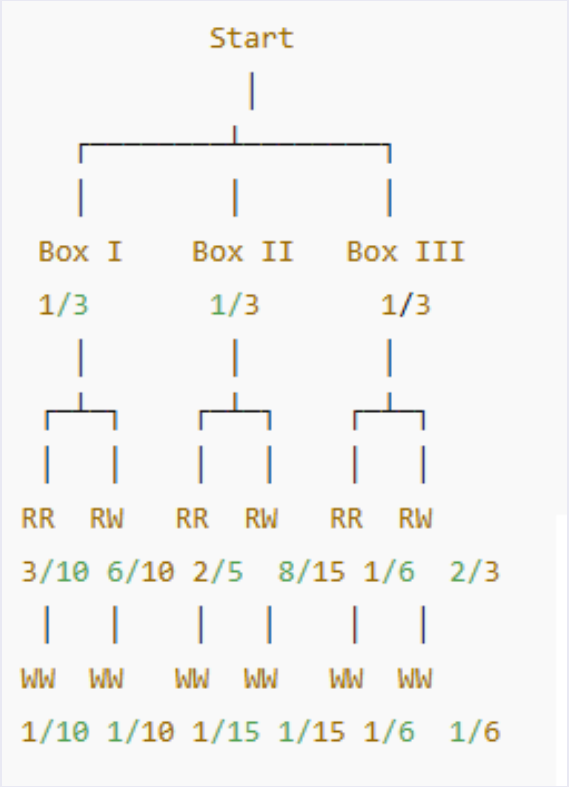
a) Draw the tree diagram.

b) What is the probability of selecting ball is red?

c) If the selected ball is red, what is the probability that head occurred on the coin?

Solution

a)



b) $P(Red) = P(HR) + P(TR) = \frac{1}{2}.\frac{3}{5} + \frac{1}{2}.\frac{2}{5} = \frac{1}{2}$

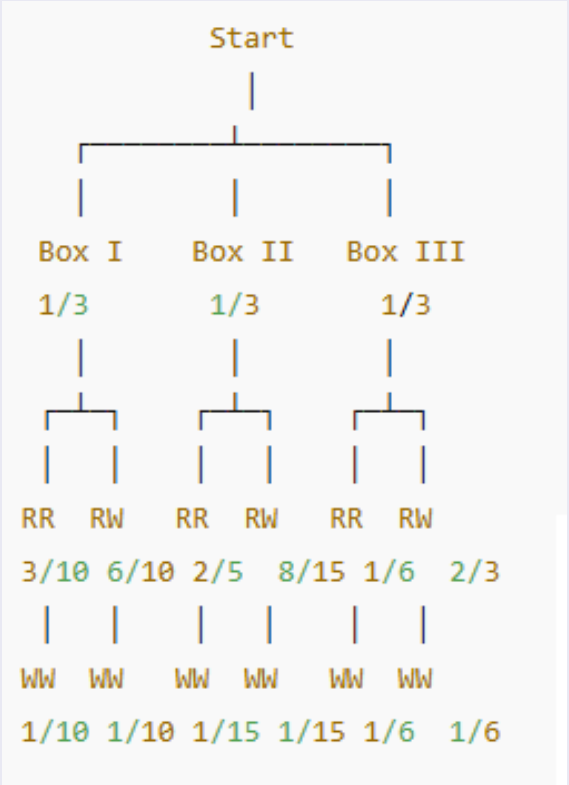
c) $P(H|R) = \frac{P(H \cap R)}{P(R)} = \frac{\frac{1}{2}.\frac{3}{5}}{\frac{1}{2}} = \frac{3}{5}$

Example

Given three boxes as follows:
Box I: 3 red and 2 white balls
Box II: 4 red and 2 white balls
Box III: 2 red and 2 white balls
First, a box is selected. Then, two balls are selected at randomly without replacement from that box.
a) Draw the tree diagram.
b) What is the probability that the selected both balls are red?
c) If the selected both balls are red, what is the probability that they are selected from Box III?

Solution

a)



b) $P(RR) = P(BoxIRR) + P(BoxIIRR) + P(BoxIIIRR) =$
$\frac{1}{3} \cdot \frac{3}{5} \cdot \frac{2}{4} + \frac{1}{3} \cdot \frac{4}{6} \cdot \frac{3}{5} + \frac{1}{3} \cdot \frac{2}{4} \cdot \frac{1}{3} = \frac{20}{36}$

c) $P(BoxIII|RR) = \frac{P(BoxIII \cap RR)}{P(RR)} = \frac{\frac{1}{3} \cdot \frac{2}{4} \cdot \frac{1}{3}}{\frac{20}{36}} = \frac{1}{10}$